INCH-POUND
MIL-DTL-19772C(SH)
03 October 2017
SUPERSEDING
MIL-DTL-19772B(SH)
09 February 1996

DETAIL SPECIFICATION

VALVES, TEMPERATURE REGULATING, AUTOMATIC, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic temperature regulating valves for naval shipboard steam, water, and other fluid service where accurate temperature control of fluid is required. These valves are of the self-operated type requiring no external power supply such as oil, water, steam, electricity, or compressed gas for their operation. However, when the failsafe mode is required (see 6.2), it can be externally powered.

1.2 Classification. Automatic temperature regulating valves are of the following types, classes, and series as specified (see 6.2):

a. Type I – Direct operated 3-way valves – series 100, 150, 250.

b. Type II – Direct operated 2-way valves.

(1) Class 1 – Direct acting (heating service) (see 6.6.4) – series 100, 150.

(2) Class 2 – Reverse acting (cooling service) (see 6.6.10) – series 100, 150, 250, 700.

c. Type III – Pilot operated 2-way valves.

(1) Class 1 – Direct acting (heating service) – series 100, 150, 400, 600.

(2) Class 2 – Reverse acting (cooling service) – series 100, 150, 400, 600.

d. Type IV – Pilot operated, 2-way, combined temperature and pressure (heating service) – series 100, 150, 400, 600 --- 4:1 ratio and 1:1 ratio.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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1.3 Part or identifying number (PIN). PINs are for general information only. The acquisition requirements (see 6.2) take precedence. The PINs to be used for valves acquired to this specification are created as follows:

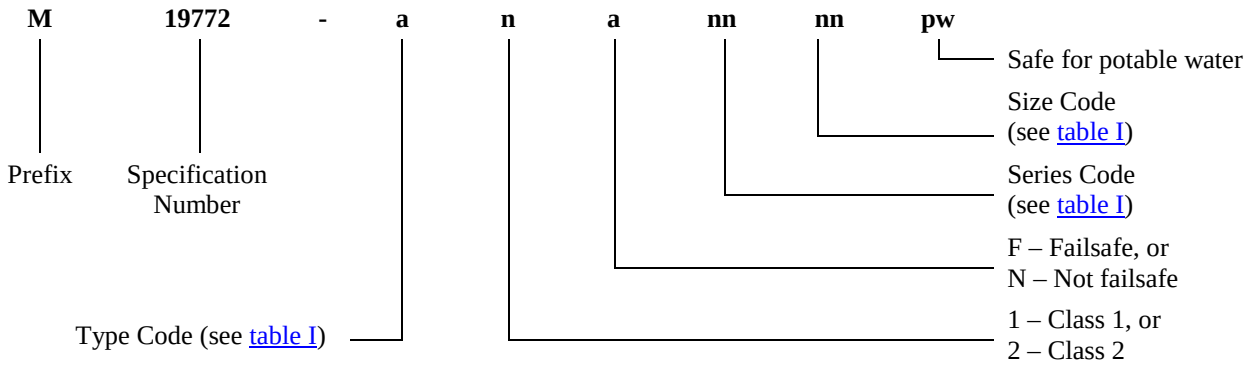


TABLE I. Code designations for PIN.

Type	Code
I	A
II	B
III	C
IV	D
Series	**Code**
100	10
150	15
250	25
400	40
600	60
700	70
Series (n.p.s.) (inches)	**Code**
¼	04
⅜	06
½	08
¾	12
1	16
1¼	20
1½	24
2	32
2½	40
3	48
3½	56
4	64
Example: The PIN for a type I, class 2, failsafe, series 100, 4-inch valve would be M19772-A2F1064.	

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver Brazing, General Specification for

MIL-PRF-2765 - Rubber Sheet, Strip, Extruded, and Molded Shapes, Synthetic, Oil Resistant

MIL-R-17131 - Rods and Powders, Welding, Surfacing

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-C-24707 - Castings, Ferrous, General Specification for

MIL-C-24707/1 - Castings, Ferrous, for Machinery and Structural Applications

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

NAVSHIPS 810-1385861 - Flanges, Sea Water, 700 PSI Max

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.12	-	Class 5 Interference-Fit Thread
ASME B16.5	-	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24 Metric/Inch Standard

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A216/A216M	-	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A313/A313M	-	Standard Specification for Stainless Steel Spring Wire
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM F1565	-	Standard Specification for Pressure-Reducing Valves for Steam Service

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-J1926/1	-	Connections for General Use and Fluid Power – Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing – Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Automatic temperature regulating valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

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3.2 Materials. Materials shall be as specified in table II and 3.2.1 through 3.2.2. Materials shall be selected to prevent galling, seizing, or excessive wear between parts. Clearance shall be such as to prevent interference due to thermal expansion. The Safe Drinking Water Act (SDWA) has stringent lead limitations on surfaces in contact with drinking water, which alloys C90300 or C92200 may not meet. Actual alloys used for valves used for drinking (potable) water system shall meet the definition of "lead free" in accordance with the requirements of the SDWA. This list of materials is not intended to be restrictive, providing it is demonstrated that proposed alternate materials are satisfactory. NAVSEA approval is required for the use of any alternate materials (see 6.2).

TABLE II. List of materials.

Name of parts	Nonferrous (series 100 and 250)	Nonferrous (series 700)	Ferrous (series 150, 400, and 600)
Body, bonnet, and pilot body (where separate)	Cast bronze ASTM B61	70 Ni-30 Cu alloy in accordance with QQ-N-281, class A or ASTM A494/A494M	Carbon steel MIL-C-24707 and MIL-C-24707/1 grade WCB of ASTM A216/A216M
Springs (external)	Carbon steel – zinc plated		
Springs (internal)	Nickel-copper alloy (UNS N04400 and N04405), chromium-nickel alloy (ASTM A313/A313M), nickel-chromium alloy (UNS N06625 and N07750), corrosion-resisting steel (300 series)		
Bellows (external)	Brass, bronze		
Bellows (internal)	Bronze, nickel-copper alloy		
Diaphragm (metallic)	Phosphor-bronze, nickel-copper alloy, corrosion-resisting steel (300 series)		
Diaphragm (nonmetallic)	Oil-resisting gasket in accordance with MIL-PRF-2765		
O-rings		Polytetrafluoroethylene or Fluorocarbon rubber	

3.2.1 Internal trim. The internal trim (valve stem, seat ring, and plug) for ferrous valves (series 150, 400, and 600) shall be corrosion-resisting steel. Unless otherwise specified (see 6.2), the seating and flow controlling surfaces on the seat ring and disc (main valve only) shall be hardfaced (see 3.3.11). Materials used for the internal trim of nonferrous valves (series 100 and 250) shall be as follows: for valves used in fresh water service, materials shall be bronze, corrosion-resisting steel, and nickel-copper alloy. for valves used in steam service, valve discs shall be of 400 series corrosion-resisting steel (hardened to 500 Brinell minimum) and seats shall be hardfaced unless otherwise approved (see 6.2); for valves employed in steam service, plastic may be used upon approval (see 6.2); for valves used in seawater service, material shall be nickel-copper alloy. The specification for nickel-copper alloy is not intended to limit the choice to basic Monel alloy, but rather is intended to permit the use of the most suitable grade of nickel-copper alloy for each particular application (for example, S-Monel and K-Monel would be considered to be nickel-copper alloy). The proper grades of the materials specified above, for use in both ferrous and nonferrous valves, shall be chosen to have the necessary strength, hardness, and nongalling properties to prevent distortion, galling, or excessive wear between operating parts.

3.2.2 External thermal bulb. The material for the external thermal bulb shall be copper for fresh water, copper or carbon steel for lube oil service, carbon steel or corrosion-resisting steel for fuel oil service, and nickel-copper alloy for seawater service.

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.4 Prohibited materials. The valves shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

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3.3 Construction. Valve construction shall be as specified in 3.3.1 through 3.3.20 (see 6.3).

3.3.1 Pressure-temperature ratings. The pressure-temperature ratings shall be as specified in table III.

TABLE III. Pressure-temperature ratings.

Series	Body material	Maximum allowable pressure (lb/in^2 gauge)	Maximum allowable temperature (°F)
100	Bronze	100	425
250	Bronze	250	150
150	Steel	150	500
400	Steel	400	775
600	Steel	600	775
700	70 Ni-30 Cu	700	150

3.3.2 End connections. End connections shall be integral with the body and as specified in table IV. Union end valves shall be furnished complete with union nuts, tailpieces, retaining ring, and seal ring.

TABLE IV. End connections.

Series	Flanged end 1/	Union end
	Reference documents	
100	MIL-PRF-20042	MIL-DTL-1183
250	MIL-PRF-20042	MIL-DTL-1183
150	ASME B16.5	---
400	ASME B16.5	---
600	ASME B16.5	---
700	810-1385861, type P	---

NOTE:

1/ Flange bolting shall straddle the vertical center line. When an odd number of bolts is used, the vertical center line at the top of the flange shall be straddled.

3.3.2.1 End connections. Type II and type III series 100 valves and type II series 250 valves in sizes up to and including 2 inches shall be union end. Type I valves in sizes up to and including 2 inches may be union end or flanged (see 6.2). Other valves shall be flanged. The ¼- and ⅜-inch type II valves shall be furnished with ½-inch end connections.

3.3.3 Accessibility. With the exception of type I valves, valves shall be accessible for adjustment, repair, or replacement of internal trim, including the thermal element, without requiring removal of the valve from the line.

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3.3.4 Capillary tubing. The capillary tubing shall be covered throughout the entire length with a flexible bronze, brass, or corrosion-resisting steel armor constructed so the armored tubing can be coiled to a 6-inch diameter without plastic deformation. At the point where the tubing joins the bulb and the valve, the tubing shall be reinforced with an additional layer of armor for a distance not less than 5 inches, constructed so this section can be bent to not less than a 7-inch diameter to prevent kinking or mechanical failure at this point. Once bent or coiled, the armored tubing shall hold its shape without mechanical restraint. The armor shall be smoothly finished and free from all mechanical defects. Unless otherwise specified (see 6.2), the length of the capillary tubing between the bulb and the valve body shall be 10 feet.

3.3.5 Thermal media. The media employed in the thermal element shall have no detrimental effect on the material with which it comes in contact.

3.3.6 Bellows. Bellows (see 6.6.5 and 6.6.8), where employed, shall be of a seamless type. Connections shall be hermetically sealed by interlocking and soldering, welding, or brazing. Bellows may be either single or double ply (laminated construction).

3.3.7 Bulb. The bulb shall be either of one-piece construction or shall be joined together by welding or silver brazing so that the joint is comparable to one-piece construction. The bulb shall be either completely flexible or flexible with a tip rigid for 4 to 7 inches. Bulb connection shall be flanged or threaded. Flange rating shall be as specified (see 6.2). Bulb length shall not exceed 22 inches for type I, II, and IV valves in sizes up to and inclusive of 1½ nominal pipe size (n.p.s.). For type I, II, and IV valves in sizes 2 inches and above, bulb length shall be kept to a minimum. Bulb length shall not exceed 15¾ inches for type III, class 1, series 100 valves in sizes up to and inclusive of 2 n.p.s. Bulb diameter of type III, class 1, series 100 valves shall be less than 3⁄16 inch.

3.3.8 Thermal element. All type II, III, and IV valves shall be furnished with an adjustable external thermal element (see 3.4.1 and 6.6.6). Type I valves shall be furnished with either an adjustable external thermal element or an adjustable or fixed setting internal thermal element (see 6.2 and 6.6.9). Thermal elements shall be interchangeable between temperature regulating valves of the same type, class, series, size, and manufacture. Thermal elements for type IV valves shall be completely interchangeable between all sizes of valves of a manufacturer for any given temperature range. For type III, class 1, series 100 valves, thermal elements shall be held in place by a 1-⅛-12 UN straight thread retaining nut and a seal having a combined length of 1 inch.

3.3.9 Overrange protection. Valves furnished with thermal elements whose service range does not extend over 100 degrees Fahrenheit (°F) shall withstand a temperature at least 50 °F above the upper range limit without detrimental effect to any part of the valve. Valves furnished with thermal elements whose service range exceeds 100 °F shall withstand a temperature at least 20 °F above the upper range limit without detrimental effect to any part of the valve. Type I valves furnished with internal thermal elements shall withstand a temperature of 215 °F without detrimental effect to any part of the valve.

3.3.10 Safety features.

3.3.10.1 Type I valves. Type I valves shall incorporate at least one of the following safety features:

a. Loss of the thermal charge will result in a valve plug position that directs at least 95 percent of the flow to the cooler (see 6.2).

b. Provision to manually regulate flow in the event of loss in the thermal charge.

c. Provision of at least two independent and interchangeable thermal elements, each capable of controlling the regulating valve at all times.

3.3.10.2 Type II and III valves. Where danger or injury to personnel or damage to equipment may occur in the event of an oversupply of heating steam or loss of cooling fluid, type II or III valves shall be of failsafe construction as defined in 6.6.7 (see 6.2).

3.3.10.3 Type IV valves. In the event of loss of the thermal charge, the thermal feedback signal shall be removed from the regulating valve and it shall function as a pressure reducing valve. The reduced pressure setting shall be manually adjustable with the valve under pressure.

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3.3.11 Seats. Except for type III and IV valves where seats may be separate or integral with the body, valve seats shall be replaceable. Type III or IV valves without separate seat rings shall require simultaneous qualification of an alternative design employing a separate seat ring. The alternative design shall only contain modifications in the seating region. The valve without separate seat rings shall be readily convertible to the alternate configuration via machining or allied process accomplished in the seat region and installation of alternate seat rings. For valves to be used on steam service above 225 pounds per square inch (lb/in^2) gauge, threaded-in seat rings shall be brazed circumferentially for bronze valves and seal welded circumferentially for steel valves. Where seating surfaces are hardfaced, the inlay shall be as specified in MIL-R-17131 and shall have a minimum thickness of 3⁄32 inch. Type IV valves shall have hardfaced seats.

3.3.12 Plug guiding. The stem/plug assembly guiding shall prevent binding or seizing due to lateral thrust, thermal expansion, or body distortion caused by internal pressure or the forces and moments imposed by the connecting piping, or any combination of these, and ensure proper seating. This requirement shall be maintained with interchangeable parts and under all tolerance stack-up conditions.

3.3.13 Fasteners. Flange bolting for other than end connections shall be one of the following:

a. Through bolts fitted with a nut on each end. Threads on bolts and nuts shall be UN class 2 fit.

b. Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be as specified in ASME B1.12.

c. For type III, class 1, series 100 valves, cap screws may be used in place of flanged bolting or tap-end studs. Bearing surfaces of nuts and bolts shall be finish machined. Their respective mating surfaces on the valve shall also be finish machined if not cast smooth and true.

3.3.14 Threads. Threads shall conform to FED-STD-H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Tapered pipe threads shall not be used. Except as specified, threaded lubing fittings shall be straight thread adjustable O-ring fittings designed to mate with threaded bosses as specified in SAE-J1926/1.

3.3.15 Springs. Springs shall be designed so that they will not be fully compressed during any normal operation or adjustment of the valve. The working stress shall be so that relaxation shall be not greater than 5 percent over a 1,000-hour period at the operating temperature. Spring ends shall be squared and ground.

3.3.16 Interchangeability. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance without requiring selection, fitting, or machining of any kind.

3.3.17 Set point adjustment. Means shall be provided for adjusting the temperature set point over the range specified in 3.4.1 with the valve under pressure. Means shall also be provided on type IV valves for adjusting the outlet pressure through a range not less than 75 to 125 percent of the mid-range set pressure with the installed spring, without requiring replacement of any internal parts. A locknut or other suitable device shall be provided to prevent accidental change of any adjustments.

3.3.18 Diaphragms. Where type III valves are of the diaphragm actuated type, and where the diaphragm is nonmetallic, the valve shall limit the temperature of the diaphragm to not greater than 180 °F at all times, with an ambient temperature of 110 °F.

3.3.19 Type IV valve requirements. Type IV valves shall be basically internal pilot operated pressure reducing valves as specified in ASTM F1565 with the addition of the provision, which automatically modifies the reduced pressure setting as a function of the downstream temperature by means of a temperature signal feedback circuit into the pilot section.

3.3.20 Valve envelope dimensions.

3.3.20.1 Dimensions. End-to-end, height, and width dimensions for type III, class 1, series 100 valves shall be as specified in table V. For other valves not listed, use manufacturer's standard.

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TABLE V. Type III, class 1, series 100 valve envelope dimensions.

Valve size	End-to-end	Maximum height below center	Maximum height above center	Maximum width
1	9½	9	17	14
1¼	10¾	9	17	14
1½	11¾	9	17	14
2	12¾	9	17	14
NOTE: Dimensions are in inches.				

3.4 Performance requirements.

3.4.1 Temperature range and accuracy of regulation. Valves shall maintain the controlled temperature within the limits specified in table VI. Minimum ranges for adjustable thermostatic elements and the service temperature range for which elements must be supplied shall be as specified in table VI.

TABLE VI. Temperature range and accuracy of regulation.

Type	Class	Adjustment limits (see 6.6.1) (°F minimum)	Service temperature range (°F)	Control band (°F)
I	---	Fixed	100 - 180	±5
I	---	30	100 - 180	±7.5
II	1	30	60 - 200	±7.5
II	2	30	60 - 160	±7.5
III	1 and 2	40	60 - 260	±4
IV	---	40	60 - 260	±7.5 [1/]

NOTE:

[1/] Control band (see 6.6.2) to be determined by the set pressure ratio test specified in 4.2.6.1.

3.4.2 Ambient temperature effect. Change in ambient temperature around the capillary tubing and valve from 80 to 110 °F shall not influence the controlled fluid temperature by greater than 4 °F. The valve shall continue to function when the ambient temperature at the bellows is 140 °F.

3.4.3 Dead band (type II only). Measured in percentage of valve stroke, dead band (see 6.6.3) shall be not greater than 5 percent.

3.4.4 Seat tightness. With the exception of types I and IV, valves shall provide lock-up at the control band limit. Type I valves shall have not greater than 5 percent or 6 gal/min, whichever is less leakage to the unopened outlet at the control band limit. Type IV valves shall provide positive lock-up within the dead-end build-up limits specified in 4.2.6.3.

3.4.5 Stability. Valves shall operate over the entire flow range without hunting, chattering, or creating excessive noise and vibration at any setting within the adjustment range.

3.4.6 Capacity. Unless otherwise specified (see 6.2), minimum required valve capacities shall be as specified in 3.4.6.1.

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3.4.6.1 Type I valve capacities. Minimum water capacities for type I valves with a 5-lb/in^2 drop across the valve and with all flow through the cooler port shall be as specified in table VII.

TABLE VII. Capacity for type I valves.

n.p.s. (inches)	Flow (gal/min)
½	6
¾	10
1	15
1¼	20
1½	25
2	86
2½	125
3	185
4	350
5	450
6	600

3.4.6.2 Type II, class 1 valve capacities. Minimum capacities for type II, class 1 valves based on a 35-lb/in^2 gauge steam inlet pressure and an outlet pressure of 20 lb/in^2 gauge or less, with the valve set in the mid-point of its adjustable range, and with the element sensing a temperature of 15 °F below the temperature at which the valve is set to close shall be as specified in table VIII.

TABLE VIII. Capacity for type II, class 1 valves.

n.p.s. (inches)	Flow (gal/min)
¼	40
⅜	80
½	100
¾	300
1	500
1¼	1,000
1½	1,300
2	2,000
2½	3,000
3	4,500
4	7,500

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3.4.6.3 Type II, class 2 valve capacities. Minimum water capacities for type II, class 2 valves with a 6-lb/in^2 drop across the valve and with the element sensing a temperature of 15 °F above the temperature at which the valve is set to close shall be as specified in table IX.

TABLE IX. Capacity for type II, class 2 valves.

n.p.s. (inches)	Capacity (gal/min)
¼	1.4
⅜	2.1
½	7
¾	10
1	13
1¼	20
1½	30
2	55
2½	80
3	125
4	200

3.4.6.4 Type III, class 1 valve capacities. Minimum capacities for type III, class 1 valves based on a 35-lb/in^2 gauge steam inlet pressure and an outlet pressure of 20 lb/in^2 gauge or less and with the element sensing a temperature of 6 °F below the temperature at which the valve is set to close shall be as specified in table X.

TABLE X. Capacity for type III, class 1 valves.

n.p.s. (inches)	Capacity (pounds steam per hour)
½	150
¾	300
1	500
1¼	1,000
1½	1,300
2	2,000
2½	3,000
3	4,500
4	7,500

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3.4.6.5 Type III, class 2 valve capacities. Minimum water capacities for type III, class 2 valves with a 5-lb/in^2 drop across the valve and with the element sensing a temperature of 6 °F above the temperature at which the valve is set to close shall be as specified in table XI.

TABLE XI. Capacity for type III, class 2 valves.

n.p.s. (inches)	Capacity (gal/min)
½	7
¾	10
1	20
1¼	30
1½	40
2	60
2½	110
3	150
4	230

3.4.6.6 Type IV valve capacities. Minimum capacities for type IV valves based on a 100-lb/in^2 steam inlet pressure and an outlet pressure of 55 lb/in^2 gauge or less and with the element sensing a temperature of 6 °F below the temperature at which the valve is set to close shall be as specified in table XII.

TABLE XII. Capacity for type IV valves.

n.p.s. (inches)	Saturated steam capacity (pounds per hour)
½	275
¾	510
1	850
1¼	1,520
1½	2,100
2	3,500
2½	5,000
3	7,800
3½	10,500
4	13,500

3.5 Mechanical shock and vibration. The valves shall withstand shock and vibration.

3.5.1 Shock resistance. Valves shall meet the grade A, class 1, type C shock test requirements of MIL-DTL-901 (see 4.2.7).

3.5.2 Vibration resistance. Valves shall meet the type I vibration test requirements of MIL-STD-167-1 (see 4.2.8).

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3.6 Marking.

3.6.1 Body markings. The manufacturer's name or trademark shall be cast or forged integral with the valve body. For type II, III, and IV valves, a flow arrow shall be permanently marked on the body. For type I valves, the inlet connection shall be marked with the letter "A", the bypass connection shall be marked with the letter "B," and the cooler connection shall be marked with the letter "C".

3.6.2 Identification plates. Each valve shall have an identification plate permanently attached to an exposed position on the valve that will not be covered by insulation. The identification plate shall be made of a corrosion-resistant material and shall contain the following information or a space therefor:

a. Manufacturer's name or trademark.
b. MIL-DTL-19772 and type, class, series, if of failsafe construction, and if safe for potable water systems.
c. Size and capacity.
d. Temperature range of element.
e. Body and trim material composition.
f. Manufacturer's drawing number.
g. Manufacturer's model and part number or identification.
h. Applicable manual number.
i. Space for 9-digit component identification number (CIN).

3.7 Nondestructive tests. Valves shall be capable of withstanding nondestructive testing (see 4.3.4.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.2.2 and the tests of 4.2.3 through 4.2.8.

4.2.1 Qualification test sample. The following categories shall require separate qualification testing:

a. Type I
b. Type I (fail safe construction) (see 6.6.7)
c. Type II, class 1
d. Type II, class 1 (fail safe construction)
e. Type II, class 2
f. Type II, class 2 (fail safe construction)
g. Type III, class 1
h. Type III, class 1 (fail safe construction)
i. Type III, class 2
j. Type III, class 2 (fail safe construction)
k. Type IV, 4:1 ratio
l. Type IV, 1:1 ratio

4.2.1.1 Size and series. Qualification of the size and series of the valve tested will include qualification of all other valves in the tested category for all sizes and for all series up to and including the series of the test valve. The size of the test valve shall be as approved by NAVSEA.

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4.2.2 Visual examination. A visual examination shall be made to verify conformance to the requirements of this specification not covered by tests.

4.2.3 Tests for type I valves.

4.2.3.1 Responsiveness. The thermal element shall be inserted in fluid at the temperature at which all of the flow is directed through the by-pass circuit. The temperature of the fluid acting on the thermal element shall be instantly raised by 30 °F. Under the new temperature condition, the valve shall reposition itself to flow not less than 70 percent of the inlet water to the cooler outlet within 35 seconds and 90 percent in 70 seconds. The flow diversion shall be demonstrated by calculations and testing.

4.2.3.2 Flow test. With the thermal element sensing a temperature equal to the upper limit of the control band (see 3.4.1), the water flow rate through the cooler port shall be determined with a 5-lb/in^2 pressure drop across the valve. Minimum flow capacity shall be as specified in 3.4.6.1.

4.2.3.3 Seat tightness. The thermal element shall sense a liquid with temperature once at each limit of the control band (see 3.4.1). At each limit, the valve shall be allowed to reposition itself to its equilibrium state; after which, flow through the port, which is closed, shall be not greater than the leakage limit as specified in 3.4.4.

4.2.4 Tests for type II valves.

4.2.4.1 Responsiveness.

4.2.4.1.1 Class 1 valves. With steam at the design operating pressure and temperature and a flow as specified in table VIII through the valve, the temperature at the thermal element shall be instantly raised 25 °F. The valve shall fully close within 40 seconds.

4.2.4.1.2 Class 2 valves. With cooling water at the design operating pressure and temperature and a flow of maximum capacity through the valve as specified in table IX, the temperature at the thermal element shall be instantly dropped 20 °F below the temperature; at which point, the valve is fully open. The valve shall fully close within 40 seconds.

4.2.4.2 Flow test. The valve shall be tested when set at the lower, mid-point, and upper limits of its adjustable range. At each setting, the temperature that the element is sensing shall be varied from the upper to lower limits of the control band specified in 3.4.1 and the fluid flow (heating or cooling, as applicable) shall vary from lock-up to the minimum required flow specified in 3.4.6.2 or 3.4.6.3 under the pressure conditions specified in 3.4.6.2 or 3.4.6.3.

4.2.5 Tests for type III valves.

4.2.5.1 Responsiveness.

4.2.5.1.1 Class 1 valves. With steam at the design operating pressure and temperature and a flow of maximum capacity through the valve as specified in table X, the temperature at the thermal element shall be instantly raised 20 °F above the temperature at which the valve is fully open. The time period shall begin when the recorded temperature of the thermal element reaches 20 °F above the operating temperature. The valve shall fully close within 15 seconds.

4.2.5.1.2 Class 2 valves. With cooling water at the design operating pressure and temperature and a flow of maximum capacity through the valve as specified in table XI, the temperature at the thermal element shall be instantly dropped 20 °F below the temperature at which the valve is fully open. The time period shall begin when the recorded temperature of the thermal element reaches 20 °F below the operating temperature. The valve shall fully close within 15 seconds.

4.2.5.2 Flow test. The valve shall be tested when set at the lower, mid-point, and upper limits of the adjustable range. At each setting, the temperature that the element is sensing shall be varied from the upper to lower limits of the control band specified in 3.4.1 and the fluid flow (heating or cooling, as applicable) shall vary from lock-up to the minimum required flow specified in 3.4.6.4 or 3.4.6.5 under the pressure conditions specified in 3.4.6.4 or 3.4.6.5.

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4.2.5.3 Test method for type III valves. Two tanks (tank 1 and tank 2) shall be used for the test with both tanks containing permanent thermometers that are partially submerged in a liquid medium. Liquid temperature in tank 2 shall be 20 °F higher than in tank 1. The temperature in both tanks shall be recorded when the liquid mediums are at steady state.

4.2.5.3.1 Class 1 valves. The thermal bulb and portable thermometer shall be initially immersed into tank 1 simultaneously until the portable thermometer's temperature is the same as the temperature indicated on the permanent thermometer. The thermal bulb and portable thermometer shall then be immersed into tank 2 simultaneously. When the temperature of the portable thermometer and thermal bulb achieves the permanent thermometer reading in tank 2, the stopwatch shall be started. The stopwatch shall be stopped at the valve closure and the total time shall be recorded.

4.2.5.3.2 Class 2 valves. The thermal bulb and portable thermometer shall be initially immersed into tank 2 simultaneously until the portable thermometer's temperature is the same as the temperature indicated on the permanent thermometer. The thermal bulb and portable thermometer shall then be immersed into tank 1 simultaneously. When the temperature of the portable thermometer and thermal bulb achieves the permanent thermometer reading in tank 1, the stopwatch shall be started. The stopwatch shall be stopped at the valve closure and the total time shall be recorded.

4.2.6 Tests for type IV valves. Test procedure shall be as follows.

4.2.6.1 Ratio test. For valves with a 4:1 ratio, set the pressure limiting device to deliver 60 lb/in^2 gauge. Set the temperature adjustment lower limit at 150 °F (so that 150 °F just starts to drop the reduced pressure). Subject the thermal element to 165 °F water and check that the reduced pressure drops to 0 lb/in^2 gauge. For valves with a 1:1 ratio, conduct as above, except set the pressure limiting device to deliver 15 lb/in^2 gauge and check that the reduced pressure drops to 0 lb/in^2 gauge with an increase of 15 °F.

4.2.6.2 Pressure limiting test. Drop the temperature that the temperature element is sensing at least 15 °F below the lower limit of the temperature control band and check that the reduced pressure does not exceed the limit setting.

4.2.6.3 Pressure regulation test. The pressure regulation test shall be conducted with the thermal system maintained constant under two conditions. First, at the lower control band limit (where the thermal feedback just begins to limit the reduced pressure) and, second, at approximately the mid-point of the control band (where the thermal feedback decreases the reduced pressure setting to approximately ½ the limit setting). Under each condition, vary steam flow from lock-up to maximum as specified in 3.4.6.6 by varying the downstream demand. The reduced pressure shall be maintained within the limits of ±5 percent or ±2 lb/in^2, whichever is greater. Dead-end buildup shall not exceed 10 lb/in^2 in a 1-hour period with a downstream volume not greater than the volume represented by 100 diameters of downstream pipe.

4.2.7 Shock. Valves shall be subjected to a shock test as specified in MIL-DTL-901 to determine conformance to the requirements specified in 3.5.1.

4.2.8 Vibration. Valves shall be subjected to a vibration test as specified in MIL-STD-167-1 to determine conformance to the requirements specified in 3.5.2.

4.2.9 Periodic retention of qualification. At intervals of not more than 3 years after initial qualification, inspection for retention of qualification shall be performed on automatic temperature regulating valves and shall include the examinations of 4.3.3 and the tests of 4.3.4. A total of 3 samples shall be selected randomly from the appropriate category of previously qualified sizes and series of valves produced within 1 year of the inspection. Inspection for retention of qualification shall be conducted at a laboratory acceptable to the Naval Sea Systems Command.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.3 and the tests of 4.3.4.

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TABLE XIII. Conformance inspection and tests.

Inspection/test	Requirement	Test	Sampling plan code letter
Material	3.2	1/	1/
Dimensional	3.3	4.3.3.2	B
Visual	3.6	4.3.3.1	B
Nondestructive tests	3.7	4.3.4.1	A
Pressure tests	3.3.1	4.3.4.2	A

NOTE:
1/ Material certifications or material test results shall be maintained to establish identity and traceability of materials used for each lot of valves.

4.3.1 Sampling. Sampling shall be as specified in 4.3.1.2.

4.3.1.1 Lot. All valves of the same type, class, series, size, thermal element, and adjustable range, using the same material and production processes, and offered for delivery at one time shall be considered a lot for the purpose of sampling.

4.3.1.2 Sampling for conformance inspection. As a minimum, a sample quantity from each lot of completed valves as specified in tables XIII and XIV shall be selected and inspected as specified in 4.3.3 and 4.3.4 (see 4.3.2). Sample size depends on the sampling plan code letter shown in table XIII.

TABLE XIV. Sample size for conformance inspection and tests.

Lot size	Sample size	
	Sampling plan code letter A	Sampling plan code letter B
2 to 8	All	5
9 to 15	13	5
16 to 25	13	5
26 to 50	13	5
51 to 90	13	7
91 to 150	13	11
151 to 280	20	13
281 to 500	29	16
501 to 1,200	34	19
1,201 to 3,200	42	23

4.3.2 Lot acceptance and rejection criteria. The contractor has the option of screening 100 percent of the rejected lot for the defective characteristic(s) or providing a new lot that shall be inspected in accordance with the sampling plan provided herein.

4.3.3 Examination.

4.3.3.1 Visual examination. A visual examination shall be made of the sample valves selected as specified in 4.3.1.2 to verify conformance to the requirements of this specification not covered by tests.

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4.3.3.2 Dimensional examination. A dimensional examination shall be made on the sample valves selected as specified in 4.3.1.2 to verify conformance to the approved drawings.

4.3.3.3 Rejection criteria. Failure of any valve in the sample to pass the examination specified in 4.3.2 shall be considered as a cause for rejection of the lot.

4.3.4 Tests.

4.3.4.1 Nondestructive tests. Nondestructive tests (radiography, magnetic particle, and dye penetrant) shall be conducted as specified in T9074-AS-GIB-010/271 and S9074-AR-GIB-010/278.

4.3.4.2 Pressure tests. Each sample valve selected as specified in 4.3.1.2 shall be pressure tested as specified in table XV. It should be noted that control diaphragms in type III valves are not designed to withstand hydrostatic test pressures; therefore, prior to testing, the diaphragms should be removed or the pilot parts should be plugged to prevent diaphragm rupture. Visible deformation or porosity or other signs of weakness under the hydrostatic test shall be cause for rejection.

TABLE XV. Pressure tests for strength and porosity.

Valve position	Hydrostatic pressure (lb/in^2 gauge)					
	100 series	150 series	250 series	400 series	600 series	700 series
Valve open	200	425	500	1,450	2,175	1,050

4.3.4.3 Rejection criteria. Failure of any valve in the sample to pass the tests specified in 4.3.4.1 or 4.3.4.2 shall be considered as a cause for rejection of the lot.

4.4 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use for each type of automatic temperature regulating valve aboard naval vessels listed in this specification is as follows.

6.1.1 Type I. Type I valves are intended for use in lube oil cooler, diesel engine cooling service, and other water mixing services where proportioning of water to a cooler and a by-pass is required.

6.1.2 Type II. Type II valves are intended for use with water heaters and other apparatus where steam or hot water is used to heat water and for condensate recirculation and general cooling services.

6.1.3 Type III, series 250 and 700. Type II, Series 250 and 700 valves are intended primarily for use in cooling service where seawater is the cooling fluid.

6.1.4 Type III. Type III valves are intended for services (such as water heaters) where fine control over a narrow range is required and where direct operated valves are not suitable for the pressures, temperatures, and response requirements.

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6.1.5 Type IV. Type IV valves are intended for use on water heaters, fuel oil heaters, and for any other apparatus where steam is used as the heating medium. Type IV valves are intended for use where close control is required and on services like instantaneous heaters where direct operated valves and type III valves are not suited.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and series required (see 1.2).

c. The specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. Approved alternate materials (see 3.2 and 3.2.1).

e. Body material (see 3.2), trim material (see 3.2.1), and bulb material (see 3.2.2), if known.

f. Whether a surface other than hardfacing is required for the internal trim of ferrous valves (see 3.2.1).

g. Requirement for prohibited material verification (see 3.2.4 and 4.4).

h. End connections required (see 3.3.2.1).

i. The length of the capillary tubing between the bulb and the valve body if other than 10 feet (see 3.3.4).

j. Bulb flange rating (see 3.3.7).

k. Type of thermal element required for type I valves (see 3.3.8).

l. Whether a special fail safe construction is required. For type I valves, see 3.3.10.1.a. For type II or III valves, see 3.3.10.2.

m. Minimum required valve capacity if other than specified (see 3.4.6).

n. Temperature setting.

o. Reduced pressure setting and ratio required for type IV valves.

p. Maximum and minimum inlet pressure and temperature conditions.

q. Brief description of application.

r. Packaging requirements (see 5.1).

s. Technical manual requirements (see 6.3).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. Manuals should be furnished as specified in MIL-DTL-24784/22.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 19772 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Interchangeability relationship. Current stocks of valves may be used until depleted. Future requisitions for valves should refer to the current version of this specification.

6.6 Definitions. The following definitions are applicable to this specification:

6.6.1 Adjustment limits. The range of set temperatures over which the valve can be adjusted while meeting the performance requirements.

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6.6.2 Control band. The temperature change required to move the valve plug through its full travel from the shut-off to the fully open position or vice versa. Influences such as internal friction and pressure drop forces must be taken into account when determining the control band for a particular application. "Accuracy of regulation" is the amount, under a steady state condition, by which the controlled temperature deviates from the set temperature (see 6.6.11). In cases where the valve plug does not need to move through its full travel to meet the range of load demands of a particular application, the "accuracy of regulation" will be finer than the control band of that valve.

6.6.3 Dead band. A measure of the inability of the valve to assume an identical position for the same temperature on its upward and downward travel.

6.6.4 Direct acting. A 2-way valve is direct acting when an increase in the temperature being measured causes the valve plug to move toward the seated position. Generally used in heating service.

6.6.5 External bellows. A bellows that is normally external to the body and is not in contact with the process fluid.

6.6.6 External thermal element. An element attached to the valve body by a suitable superstructure and to an externally located thermal bulb via capillary tubing.

6.6.7 Fail safe. A temperature regulating valve that incorporates a special thermal system which, in the event of leakage from the thermal element, positions the valve plug at the safe (as defined below) end of its travel:

a. 3-way valve (type I) – Valve plug position resulting from failure of the thermal element will divert at least 95 percent of the flow to the cooler.

b. 2-way valve (types II or III) – Direct acting valves used for heating to fail closed and shut off steam flow. Reverse acting valves used for cooling service to fail open to pass full cooling fluid flow.

6.6.8 Internal bellows. A bellows that is normally in contact with the process fluid.

6.6.9 Internal thermal element. An element that is entirely enclosed within the valve body that measures the temperature of the fluid passing through the valve.

6.6.10 Reverse acting. A 2-way valve is reverse acting when an increase in the temperature being measured causes the valve plug to move away from the seated position. Generally used in cooling service.

6.6.11 Set temperature. The nominal setting of the valve. Usually taken to be the mid-point of the range within which the temperature is required to be controlled or the temperature at the mid-point of the valve plug travel.

6.7 Subject term (key word) listing.

Direct acting

Fluid service valve

Reverse acting

Self-operated valve

Steam valve

Water valve

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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